Net force first

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Combine forces before using mass to predict acceleration.

SUCCESS CRITERIA - I can...

1. Find net force before dividing by mass.
2. State acceleration with units and direction.
3. Separate acceleration direction from motion direction.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “net force” and “mass” to explain your model or result. Sentence starters are optional.

net force: the vector sum of all forces on one object

mass: a measure of how strongly an object resists acceleration

acceleration: rate of change of velocity

01 Warm-up: bring back an earlier idea

A 5 kg sign hangs still from one vertical cord.
Use g=10 N/kg.
Name both forces, their sources, magnitudes and directions.

Sentence starter: I notice ... . This supports ... because ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: ΣF = ma; a = ΣF/m.
Add signed forces on the same object before dividing by mass. N/kg = m/s².

Worked reasoning: A 2 kg cart has 9 N right and 3 N left. Net force=6 N right; a=6/2=3 m/s² right. Add forces before dividing by mass.

Your turn: Draw arrows for both forces, then a separate net-force arrow.
Explain why 9/2 is not the acceleration.

Sentence starter: My claim is ... . The diagram/data show ..., so ... .



2 kg cart: draw the net force separately.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read the arrows

For the 4 kg cart, find net horizontal force and acceleration. Write each step.

Sentence starter: I found ... [unit]. This value means ... in this situation.



The dot represents the object. Mass: 4 kg

Marzano check - circle: 1 2 3

04 Balanced but moving

A 2 kg cart is already moving left when two forces act: 5 N right and 5 N left.
Use the diagram to find its acceleration.
Describe its motion if these forces remain balanced.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Marzano check - circle: 1 2 3

05 Compare two masses

Two test carts experience the same 6 N net force. Complete the comparison table. Which cart changes velocity faster, and what causes the difference?

Sentence starter: Both ... . They differ in ... because ... .

Mass (kg)	Net force (N)	a (m/s²)
2	6	____
3	6	____

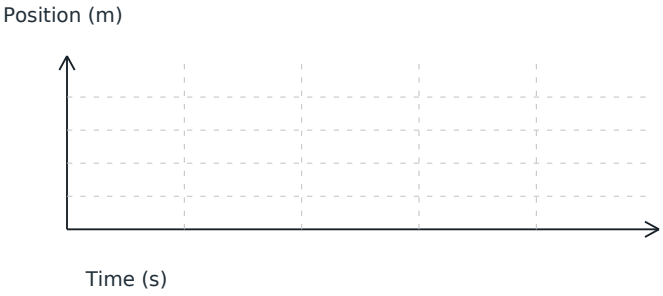
Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A cart is at 5, 12, 19 m at 0, 1, 2 s.
Predict its position at 9 s.
State an assumption and a measurement that could challenge it.
Label the model with the given positions and any prediction.

Word help: prediction = what a model says may happen next

Sentence starter: I predict ... . My model supports this because ... .



Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

An 8 kg object has 50 N right and 18 N left.
Draw the forces, calculate net force, and find a with direction.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Add and label force arrows.

Draw the given forces; show net force separately.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Find two different pairs of opposite horizontal forces that give the same 6 N net force right. Keep mass 2 kg and compare accelerations.

Sentence starter: Both ... . They differ in ... because ... .



Add and label force arrows.

Label each interaction force on the cart.

Marzano check - circle: 1 2 3